

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Seventh Semester B.E. (Credit System) Degree Examinations

January 2023

19EE8X10 – NON-CONVENTIONAL ENERGY SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Give a brief classification of energy resources. Explain the role of energy consumption in the prosperity of a nation.	10	L*2	1	1,7
	b) Discuss the status of present scenario of both India and world. Mention the salient features of non-conventional energy sources and their limitations.	10	L2	1	1,7
2.	a) Define the following terms in connection with solar radiation. Solar constant; Angle of incidence; Zenith angle; Beam radiation; Surface azimuth angle;	10	L1	2	1,7
	b) Estimate the monthly average of daily global radiation on a horizontal surface at a place with latitude and longitude of 27°10' N and 78°05' E (angle of latitude = 27.166°) respectively on the 17th of January if the average sunshine hour per day is 7. Take a=0.25 and b=0.57	10	L5	2	1,7
3.	a) With the help of neat sketch explain the construction and working of Solar flat plate collectors and Solar furnace.	10	L2	2	1,7
	b) Explain the structure of typical stand-alone solar photovoltaic system. Differentiate between the solar cell, solar PV module and solar PV array.	10	L2	2	1,7
Unit – II					
4.	a) With the proper schematic representation, explain the working of the following: Compressed air energy storage and Energy storage using flywheels	10	L2	3	1,7
	b) Discuss the need for storage of energy in renewable energy-based systems. Mention some of the important types of energy storage systems.	10	L2	3	1,7
5.	a) Explain the basic principle of working of wind electric conversion system. Derive an expression for available energy from wind.	10	L2	3	1,7
	b) Draw the neat sketch of horizontal axis wind turbine and explain the functions of different components.	10	L2	3	1,7
6.	a) Explain the power versus wind speed characteristics of a typical wind turbine. A propeller type wind turbine has following data: Wind speed at a height of 10 m = 12 m/s Air density = 1.226 kg/m ³ Surface roughness constant=0.14°				

P.T.O.

Height of the tower = 100 m

Diameter of the rotor = 80 m

Reduction in the wind velocity at turbine = 20%

Generator efficiency = 85%

Find

- | | | | | |
|---|----|----|---|-----|
| i) Total wind power available | 10 | L5 | 3 | 1,7 |
| ii) Power extracted by turbine and | | | | |
| ii) Electrical power generated | | | | |
| Take interference factor of 0.2 and power coefficient $C_p = 0.512$ | | | | |
| b) Discuss the various factors affecting the operation of a biogas plant. Mention some of the key usable forms of biomass, their composition, and properties. | 10 | L2 | 4 | 1,7 |

Unit – III

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|---|----|----|---|-----|
| 7. a) Discuss the principle behind the generation of power from tides. Compare the performance of single and double basin type of tidal power plants. | 10 | L2 | 5 | 1,7 |
| b) Explain the working principles of open cycle type ocean thermal energy conversion plant. Mention the environmental impacts due to the ocean thermal energy conversion. | 10 | L2 | 5 | 1,7 |
| 8. a) Compare the operation and performance of conventional and micro hydro power stations. Explain the working of a typical micro hydro power station with the help of a layout diagram. | 10 | L2 | 5 | 1,7 |
| b) Write the note on the following. | | | | |
| i) Fuel cells ii) Wave energy | 10 | L2 | 5 | 1,7 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
